Class Activity: Transcription and Translation in Biopython

Activity Title: DNA → RNA → Protein: Understanding Transcription & Translation

Learning Objectives:

By the end of this activity, students should be able to:

- Understand the **biological process** of transcription (DNA → mRNA) and translation (mRNA → Protein).
- Implement **Biopython functions** to transcribe and translate DNA sequences.
- Apply their knowledge to solve real-world **sequence analysis exercises**.

Activity Overview

This class activity consists of **three parts**:

1. **Concept Explanation** – Understanding transcription and translation.
2. **Hands-on Coding Exercise** – Writing Python scripts using Biopython.
3. **Group Challenge & Exercises** – Solving problems in teams and discussing results.

Part 1: Concept Explanation (10 minutes)

1. Transcription: DNA → mRNA

- In cells, DNA is **transcribed** into **mRNA** (messenger RNA).
- **Base-pair rules for transcription:**
    - **A (Adenine) → U (Uracil)**
    - **T (Thymine) → A (Adenine)**
    - **C (Cytosine) → G (Guanine)**
    - **G (Guanine) → C (Cytosine)**

Example:

DNA	T	A	C	G	G	A	T	C	A
mRNA	A	U	G	C	C	U	A	G	U

---

2. Translation: mRNA → Protein

- The mRNA sequence is **translated** into a **protein** using **codons** (3-letter nucleotide sequences).
- Each codon represents a specific **amino acid**.
- The translation starts at **AUG** (start codon) and stops at **UAA, UAG, or UGA** (stop codons).

Example Codon Table (Partial):

Codon	Amino Acid
AUG	Methionine (Start)
UUU	Phenylalanine
UGC	Cysteine
GGA	Glycine
UGA	Stop (End of Translation)

Example:

mRNA	AUG	GCC	UAC	UGG	UGA
Protein	Methionine	Alanine	Tyrosine	Tryptophan	STOP

Part 2: Hands-on Coding Exercise (20 minutes)

1. Transcribing DNA to mRNA

Instructor Demo

```
from Bio.Seq import Seq

# DNA sequence
dna_seq = Seq("ATGCGTACGTTAG")
```

```
# Transcribe to mRNA
mRNA = dna_seq.transcribe()
print("mRNA Sequence:", mRNA)
```

2. Translating mRNA to Protein

Instructor Demo

```
# Translate mRNA to protein
protein = mRNA.translate()
print("Protein Sequence:", protein)
```

3. Full Pipeline: DNA → mRNA → Protein

Student Activity

Modify the code to:

- Take **user input for a DNA sequence**.
- Perform **transcription and translation**.
- Handle **errors (e.g., invalid bases)**.

Part 3: Group Challenge & Exercises (20 minutes)

🚀 Group Challenge (10 minutes)

1. **Break into small groups** (2-3 students per group).
2. Each group will be given **a different DNA sequence**.
3. They need to **transcribe and translate** the sequence.
4. **Identify start and stop codons** and explain the resulting protein sequence.

Example DNA Sequences for Groups

- **Group 1:** "ATGGCCATTGTAATGGGCCGCTGAAAGGGTGCCCGATAG"
- **Group 2:** "ATGTTTCCCGGGAATCTAA"
- **Group 3:** "ATGGGTTGAGTAA"
- **Group 4:** "ATGAGGGTCCCTGA"

After **10 minutes**, each group presents their results!

📝 Individual Exercises (10 minutes)

Exercise 1: Find the Start Codon

Modify the given DNA sequence to:

- Check **if a start codon (ATG) is present**.
- If found, **transcribe and translate** it.
- If **not found**, print "No start codon found".

Exercise 2: Extract All Codons

Write a function to:

- Read an mRNA sequence.
- **Extract all codons** (sets of 3 nucleotides).
- Print them **line by line**.

Exercise 3: Identify Stop Codon

Modify the translation code to:

- **Stop translation at the first stop codon** (`UAA, UAG, UGA`).

Expected Learning Outcomes

- Understand the **biological concepts** of transcription and translation.
- Implement **Python scripts** to transcribe and translate sequences.
- Work in teams to **solve sequence analysis problems**.
- Identify **start and stop codons** in a DNA sequence.

Additional Resources

- **Biopython Documentation:** https://biopython.org/wiki/Documentation
- **NCBI Resources:** https://www.ncbi.nlm.nih.gov/